

Module 15: Population and Systems Ecology — Keys to Success

Learning Objectives

By the end of this module, you should be able to:

1. Diagram and distinguish between exponential and logistic population growth models.
2. Define carrying capacity and identify limiting factors.
3. Trace the path of energy through an ecosystem via trophic levels.
4. Apply the 10% Rule of Energy Transfer.
5. Explain how chemicals (like Carbon and Nitrogen) cycle through the biosphere.

The Big Picture

Evolution happens on the stage of Ecology. Organisms don't just exist in a vacuum; they eat, are eaten, excrete waste, and compete for limited space. In ecology, there are two golden rules: **Energy flows** (it enters from the sun, moves through life, and exits as heat), but **Matter cycles** (there is no "new" carbon being made; you are made of the exact same carbon atoms that once made up dinosaurs).

Key Terms & Concepts

1. Population Growth

Populations grow as individuals reproduce, but they do so in recognizable mathematical patterns:

- **Exponential Growth (J-Curve):** A population grows constantly at an accelerating rate. The graph looks like a steep "J". This happens when a population has **unlimited resources** (e.g., bacteria in a fresh petri dish, or humans since the industrial revolution). It cannot last forever.

- **Logistic Growth (S-Curve):** A population grows rapidly at first, but then levels off as it runs out of resources. The graph flattens out to look like an "S".

2. Carrying Capacity (\$K\$)

The flat top of the S-curve is the **Carrying Capacity (\$K\$)**. This is the maximum population size that a specific environment can sustain over the long term. - It is determined by **Limiting Factors**. - *Density-dependent factors*: Worsen as the population gets crowded (disease, competition for food, predators). - *Density-independent factors*: Affect a population regardless of its size (wildfires, floods, extreme winter).

3. Trophic Levels & The 10% Rule

Energy enters almost every ecosystem as sunlight and is converted into sugar by plants. It then moves up the food chain.

- **Producers (Autotrophs):** Plants, algae, phytoplankton. They make their own food.
- **Primary Consumers:** Herbivores (eat producers).
- **Secondary Consumers:** Carnivores (eat primary consumers).
- **Tertiary Consumers:** Apex predators (eat secondary consumers).

The 10% Rule: As energy moves from one trophic level up to the next, approx. **90% of it is lost** (used for moving, breathing, or simply radiated as body heat). Only **10%** of the energy actually becomes physical body mass that the next level can eat. - *Example:* If producers capture 10,000 joules of sunlight energy, primary consumers only get 1,000, secondary get 100, and tertiary get exactly 10. - *Why it matters:* This is why there are millions of blades of grass, thousands of zebras, but only a handful of lions in an African savanna. The ecosystem geometrically cannot support that many top predators.

4. Biogeochemical Cycles

While energy flows through and dissipates into space, chemical elements recycle endlessly.

- **The Carbon Cycle:** Driven heavily by the twin engines from Module 06: **Photosynthesis** (pulls Carbon Dioxide out of the air to make solid sugar) and **Cellular Respiration** (burns sugar and breathes Carbon Dioxide back into the air).

- **The Nitrogen Cycle:** Nitrogen makes up 78% of the air we breathe, but plants and animals can't use it in gas form. Certain crucial soil bacteria "fix" nitrogen (pull it from the air into solid form) so plants can absorb it by their roots. Without these bacteria, all plant and animal life would starve for nitrogen (which is needed to build DNA and Proteins!).

Study Tips

1. **Energy arrows point TO the eater.** When looking at a food web, arrows show the flow of energy. Grass $\rightarrow$ Cow means the energy from the grass goes into the cow.
2. **Remember your cellular processes.** The carbon cycle is literally just chloroplasts (photosynthesis) and mitochondria (respiration) working back and forth on a planetary scale.